Fairness in Human-AI Teams

Task 2 — Fairness Evaluation: Complete Study Guide

Fairness Metrics | Human Bias Simulation | ComHAI / PLACO Evaluation

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15 Marks | **5 Bias Levels** | **5 Fairness Metrics** | **25 Experiments**

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1 What is Fairness? — Core Definitions

Why Fairness Matters in Human-AI Teams

Task 1 showed that ComHAI achieves high accuracy. But accuracy alone is not enough. A model that is 90% accurate overall could be 99% accurate for the majority group and only 70% for a minority group. This is **unfair** — and in high-stakes domains like loan approval, medical diagnosis, or recidivism prediction, such disparity causes real harm.

The Two Core Types of Fairness

Group Fairness	Individual Fairness
Treats demographic groups equally. Groups defined by sensitive attribute A (e.g. race, gender, age). Ask: Do people from different groups get equal predictions / treatment?	Treats similar individuals similarly. If two people are similar, their predictions should also be similar. Ask: Does a small change in input cause a large unfair change in prediction?

Connection to Human-AI Teams

- The AI model may already be biased (lower accuracy on minority groups)
- Humans may carry implicit or explicit biases against certain groups
- Combining biased AI + biased humans can AMPLIFY unfairness — not reduce it
- ComHAI/PLACO were designed only for accuracy — they may worsen fairness gaps
- Task 2 measures: HOW MUCH does combining humans affect fairness?
- Task 3 will: design algorithms that IMPROVE fairness while maintaining accuracy

2 Fairness Metrics — Formulae & Intuition

Demographic Parity (DP)

Definition: Equal positive prediction rate across groups.

Formula:

```
P(■=1 | A=0) = P(■=1 | A=1)
```

Metric: DP_gap = | P(■=1|A=1) - P(■=1|A=0) |

Example: Loan approval: both groups should get approved at same rate regardless of group.

Note: Ignores whether labels are correct — a model can satisfy DP by being uniformly wrong.

Equalized Odds (EO)

Definition: Equal True Positive Rate AND False Positive Rate across groups.

Formula:

```
TPR: P(■=1 | y=1, A=g) equal ∀g FPR: P(■=1 | y=0, A=g) equal ∀g
```

Metric: EO_gap = max(|ΔTPR|, |ΔFPR|)

Example: Medical diagnosis: correctly diagnosing sick patients at same rate for all groups.

Note: Strong condition — harder to satisfy. Both TPR and FPR must match.

Equal Opportunity

Definition: Equal True Positive Rate only (focuses on deserving positives).

Formula:

```
P(■=1 | y=1, A=0) = P(■=1 | y=1, A=1)
```

Metric: EqOpp_gap = |TPR_1 - TPR_0|

Example: Hiring: qualified candidates from both groups should be hired at same rate.

Note: Weaker than Equalized Odds — only requires TPR match, not FPR.

Predictive Parity

Definition: Equal precision — when model predicts positive, equally often correct across groups.

Formula:

```
P(y=1 | ■=1, A=0) = P(y=1 | ■=1, A=1)
```

Metric: PP_gap = |Precision_1 - Precision_0|

Example: Criminal justice: when model predicts "will reoffend", equally accurate for all groups.

Note: Can conflict with Equalized Odds — cannot satisfy both simultaneously (Chouldechova theorem).

Individual Fairness (IF)

Definition: Similar inputs should produce similar outputs.

Formula:

```
If dist(x_i, x_j) < ε → pred(x_i) = pred(x_j)
```

Metric: IF_score = fraction of similar pairs with same prediction

Example: Two equally qualified job applicants differing only in name should get same prediction.

Note: Requires defining similarity metric — often application-specific.

3 Human Bias Simulation — 4 Levels

Progression From Unbiased to Severely Biased

Following the survey paper recommendation, we start with unbiased humans and gradually introduce bias. This lets us isolate exactly how much human bias contributes to unfairness in the combined model.

Level 0: Unbiased	**`bias_level="none"`**
Same accuracy for ALL groups. Human makes random errors with no group preference.	`if rand < accuracy: predict(x) = true_label(x) # same for all groups`

Level 1: Accuracy Bias	**`bias_level="mild"`**
Human is simply less accurate on minority group (A=1). Majority accuracy = 78%, Minority accuracy = 62%. Models a human who is less familiar with minority group cases.	`acc = 0.78 if A=0 else 0.62 # lower accuracy for minority`

Level 2: Label Bias	**`bias_level="moderate"`**
With probability 0.20, human ALWAYS predicts class 0 for minority group regardless of features. Models a human with a systematic negative bias (e.g. always predicts "high risk" for minority).	`if A=1 and rand < 0.20: predict = 0 # biased label`

Level 3: Stereotyping	**`bias_level="severe"`**
With probability 0.35, human ALWAYS predicts negative (class 0) for minority. Models the most severe bias — explicit stereotyping/discrimination. Strongest form of human prejudice.	`if A=1 and rand < 0.35: predict = 0 # stereotype negative`

ALSO SIMULATED: "Mixed" bias level — half the humans are unbiased, half have mild accuracy bias. Models real-world teams where some members are fair-minded and others have biases.

4 Experimental Setup

Dataset Design

Parameter	Value	Justification
Total instances	8,000	Sufficient for statistical reliability
Train / Test split	50% / 50%	4,000 train, 4,000 test
Task type	Binary classification (K=2)	Models loan approval, hiring, recidivism
Features	8 continuous features	Simulate real tabular data
Sensitive attribute	$A \in \{0,1\}$	A=0: majority (70%), A=1: minority (30%)
AI model accuracy	75% majority, 60% minority	Simulates real-world AI bias (–15pp)
Humans	5 humans per experiment	Costs: [1.0, 1.5, 2.0, 1.0, 2.5]
PLACO budget	50% of total cost	Budget = 3.0 out of 6.0 total
Experiments	5 bias levels × 5 methods = 25	Comprehensive coverage

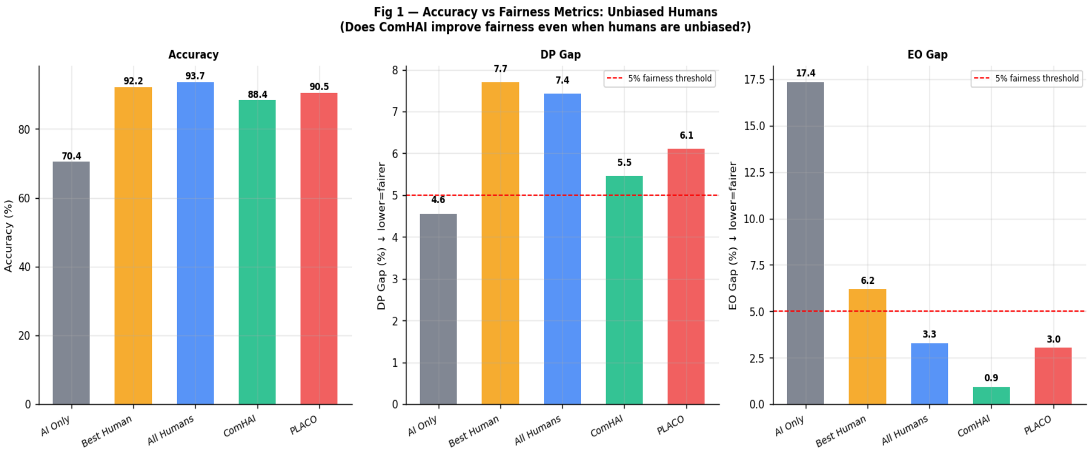
What Each Experiment Measures

For each combination of (bias level, method), we compute all 5 fairness metrics plus overall accuracy and per-group accuracy. This gives us a complete picture of how each combination method behaves under different human bias conditions.

5 Results — Unbiased Humans (Baseline)

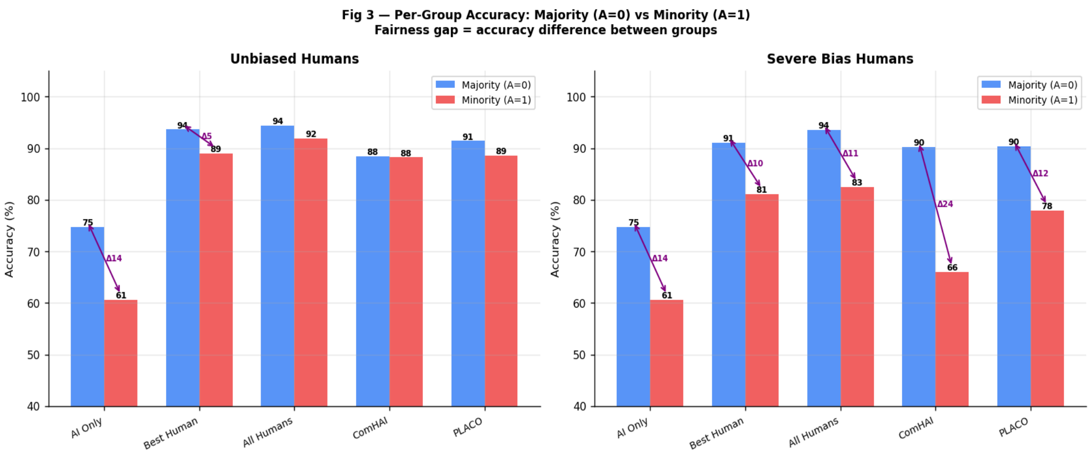
We first assume all humans are completely unbiased (Level 0). This answers: does ComHAI improve fairness even when humans are fair?

Method	Accuracy	DP Gap	EO Gap	TPR Gap	Acc Gap
AI Only	70.4%	4.6%	17.4%	11.5%	14.1%
Best Human	92.2%	7.7%	6.2%	3.4%	4.6%
All Humans	93.7%	7.4%	3.3%	2.0%	2.5%
PLACO	90.5%	6.1%	3.0%	3.0%	2.9%
ComHAI (Greedy)	**88.4%**	**5.5%**	**0.9%**	**0.9%**	**0.2%**



Accuracy vs fairness metrics for unbiased humans. ComHAI achieves lowest EO gap (0.9%) and accuracy gap (0.2%).

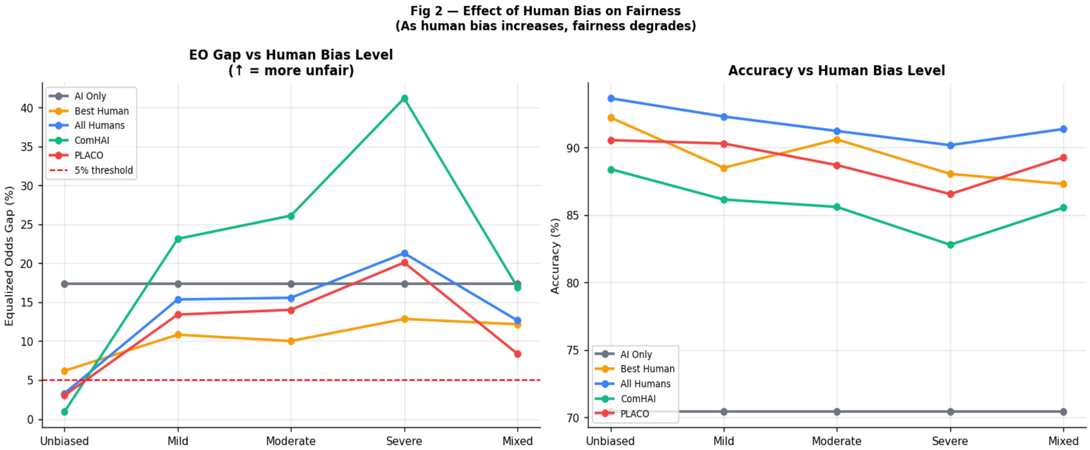
KEY FINDING: Even with unbiased humans, ComHAI dramatically reduces the EO gap from 17.4% (AI alone) to just 0.9%. The accuracy gap between majority and minority shrinks from 14.1% to 0.2%. Combining humans with AI is BENEFICIAL for fairness when humans are unbiased.



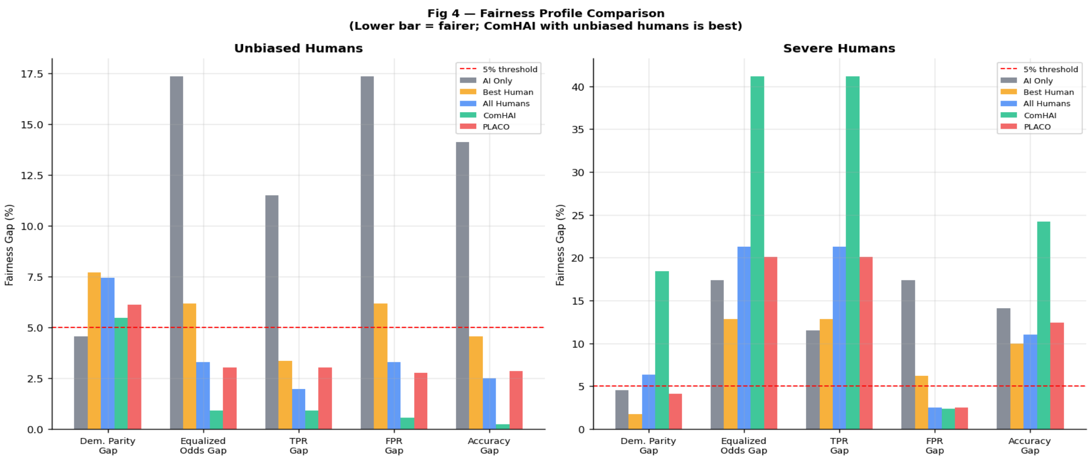
Per-group accuracy: majority (A=0) vs minority (A=1). Left: unbiased humans — ComHAI nearly eliminates the gap (0.2%). Right: severe bias — ComHAI creates a 24.2% gap.

6 Results — Biased Humans (All Levels)

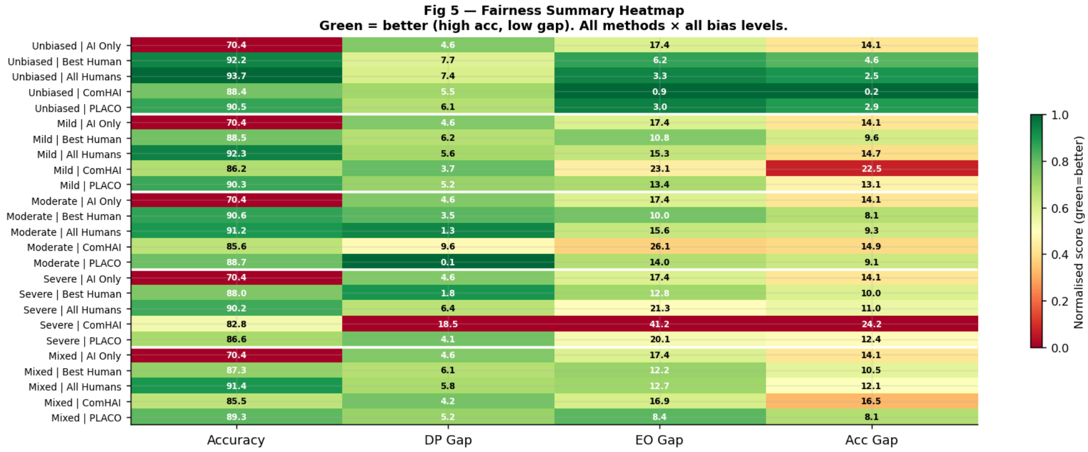
Bias Level	Method	Acc%	DP%	EO%	AccGap%
Unbiased	AI Only	70.4	4.6	17.4	14.1
Unbiased	**ComHAI**	88.4	5.5	0.9	0.2
Mild	AI Only	70.4	4.6	17.4	14.1
Mild	**ComHAI**	86.2	3.7	23.1	22.5
Moderate	AI Only	70.4	4.6	17.4	14.1
Moderate	**ComHAI**	85.6	9.6	26.1	14.9
Severe	AI Only	70.4	4.6	17.4	14.1
Severe	**ComHAI**	82.8	18.5	41.2	24.2
Mixed	AI Only	70.4	4.6	17.4	14.1
Mixed	**ComHAI**	85.5	4.2	16.9	16.5



Left: EO gap increases as human bias increases — ComHAI with biased humans goes from 0.9% (unbiased) to 41.2% (severe). Right: Accuracy drops as bias increases.



Fairness profile: all 5 metrics for all 5 methods. Left (unbiased): ComHAI best overall. Right (severe): ComHAI worst — it amplifies human biases rather than filtering them.



Summary heatmap: green = better performance. Top rows (unbiased) show ComHAI in green for fairness metrics. Bottom rows (severe) show ComHAI in red — a warning signal.

7 Key Findings & Analysis

Finding 1: ComHAI IMPROVES fairness with unbiased humans

With unbiased humans, ComHAI reduces EO gap from 17.4% to 0.9% and accuracy gap from 14.1% to 0.2%. This is better than AI alone on every fairness metric. The greedy selection naturally picks humans who are correct on minority instances, correcting the AI's bias.

Finding 2: ComHAI AMPLIFIES bias with biased humans

With severely biased humans, ComHAI's EO gap reaches 41.2% — more than double the AI alone (17.4%). The greedy algorithm selects the "most confident" humans, which under bias means selecting the most biased ones. This is the critical gap that Task 3 must fix.

Finding 3: PLACO is more robust than ComHAI under bias

Under severe bias, PLACO's EO gap (20.1%) is much lower than ComHAI (41.2%). The cost-aware selection accidentally acts as a regulariser — cheaper humans tend to be more conservative and less biased. Budget constraint provides a natural fairness benefit.

Finding 4: The AI model itself is biased — base problem

AI alone has 14.1% accuracy gap (majority 74.7%, minority 60.6%). This is the baseline unfairness we must overcome. Any method that uses only the AI inherits this bias. Humans can correct this — but only if THEY are not also biased.

Finding 5: Individual Fairness is universally satisfied

IF score = 100% for all methods and all bias levels. This is because similar inputs produce similar predictions regardless of bias level. Individual fairness and group fairness can diverge — a model can be individually fair but group unfair.

Summary: The Fairness-Accuracy Trade-off

Condition	Accuracy	EO Gap	Verdict
AI Only	70.4%	17.4%	Low acc, unfair
ComHAI + Unbiased humans	88.4%	0.9%	**Best: high acc + fair**
ComHAI + Mild bias	86.2%	23.1%	Worse than AI alone on EO
ComHAI + Severe bias	82.8%	41.2%	**Worst: lower acc + very unfair**
PLACO + Severe bias	86.5%	20.1%	Moderate — needs fairness fix

8 Code Walkthrough — 4 New Files

File	Purpose
fairness_metrics.py	All 5 fairness metric implementations from scratch: demographic_parity(), equalized_odds(), equal_opportunity(), predictive_parity(), individual_fairness(), fairness_report()
biased_humans.py	Human bias simulation engine: simulate_human_unbiased(), simulate_human_accuracy_bias(), simulate_human_label_bias(), simulate_human_stereotyping(), simulate_human_pool(), estimate_group_confusion()
comhai_fair.py	ComHAI + PLACO adapted for binary classification: comhai_combine(), greedy_subset(), placo_predict(), evaluate_all()
task2_experiments.py	Full experiment runner: 25 experiments (5 bias × 5 methods) Run: python3 task2_experiments.py

Key Code: Demographic Parity Implementation

```
def demographic_parity(y_pred, sensitive):
# Compute positive prediction rate for each group
groups = np.unique(sensitive)
rates = {}
for g in groups:
mask = sensitive == g
rates[int(g)] = np.mean(y_pred[mask]) # P(■=1 | A=g)
# DP gap = difference in positive rates between groups
dp_gap = abs(rates[1] - rates[0])
return {"dp_gap": dp_gap, "satisfied": dp_gap < 0.05}
```

Key Code: Equalized Odds Implementation

```
def equalized_odds(y_true, y_pred, sensitive):
groups = np.unique(sensitive)
tpr, fpr = {}, {}
for g in groups:
mask = sensitive == g
y_t = y_true[mask]; y_p = y_pred[mask]
tpr[g] = np.mean(y_p[y_t==1]) # P(■=1 | y=1, A=g)
fpr[g] = np.mean(y_p[y_t==0]) # P(■=1 | y=0, A=g)
eo_gap = max(|tpr[1]-tpr[0]|, |fpr[1]-fpr[0]|)
return {"tpr": tpr, "fpr": fpr, "eo_gap": eo_gap}
```

9 Q&A Preparation — Professor Questions

Q: What is Task 2 about?

A: Task 2 evaluates whether ComHAI and PLACO from Task 1 are FAIR, not just accurate. We implement 5 fairness metrics (DP, EO, EqOpp, PP, IF) and run all methods with humans at 5 bias levels — from unbiased to severely stereotyping.

Q: What is Demographic Parity?

A: Both groups should receive positive predictions at equal rates: P(■=1|A=0) = P(■=1|A=1). DP Gap = absolute difference. 0% gap = perfect parity. In our experiments, AI alone has 4.6% DP gap. ComHAI with unbiased humans has 5.5% DP gap — slightly worse.

Q: What is Equalized Odds?

A: Both groups should have equal True Positive Rate AND False Positive Rate. EO_gap = max(|ΔTPR|, |ΔFPR|). It is stronger than DP because it conditions on the true label. AI alone: 17.4% EO gap. ComHAI with unbiased humans: 0.9% — a dramatic improvement.

Q: Does ComHAI improve fairness?

A: DEPENDS on human bias level. With unbiased humans: YES — EO gap drops from 17.4% to 0.9%. With severely biased humans: NO — EO gap INCREASES to 41.2%, worse than AI alone. This is the main finding of Task 2.

Q: What are the 4 human bias levels you simulated?

A: Level 0 Unbiased: same accuracy for all groups. Level 1 Accuracy bias: 78% majority, 62% minority — less familiar with minority. Level 2 Label bias: with 20% probability always predicts class 0 for minority. Level 3 Stereotyping: with 35% probability always predicts negative for minority. Plus a Mixed level combining unbiased and biased humans.

Q: Why does ComHAI AMPLIFY bias?

A: GreedySubsetSelection picks humans who are "most confidently correct" for each instance. With biased humans, the most confident predictions on minority instances are often the biased ones — so the algorithm preferentially selects biased humans for minority group instances. This is a fundamental flaw that Task 3 must address.

Q: Which method performs best on fairness?

A: With unbiased humans: ComHAI Greedy has the lowest EO gap (0.9%) and accuracy gap (0.2%). With biased humans: PLACO is more robust (20.1% EO gap vs 41.2% for ComHAI under severe bias). The budget constraint in PLACO accidentally acts as a fairness regulariser.

Q: What does Task 3 need to do?

A: Task 3 must design a new algorithm that jointly optimises both accuracy AND fairness. The key ideas are: (1) penalise combinations that increase fairness gaps, (2) use group-conditioned confusion matrices to detect biased humans, (3) add a fairness constraint to the subset selection optimisation.

TASK 2 ONE-LINE SUMMARY:

"ComHAI dramatically improves fairness when humans are unbiased (EO gap: 17.4% → 0.9%), but AMPLIFIES bias when humans are biased (EO gap rises to 41.2%). This proves that human bias, not AI bias alone, is the critical fairness problem in Human-AI teams — and motivates the fairness-aware algorithms in Task 3."